



Project Report — Ripple, an Agentic CRM Platform

An AI layer that turns a passive CRM into an active sales teammate.

WHAT WE BUILT

Ripple is an agentic AI layer built on top of **Twenty**, an open-source CRM that serves as the system of record and the user interface. A standard CRM is passive — it stores what reps remember to type in, and goes stale the moment they stop. Ripple makes it active. On the Twenty foundation we built three capabilities that work together:

- **A conversational CRM agent.** Reps read and modify CRM data in plain language. An orchestrator interprets each request, plans it into ordered steps, and delegates to specialised read and write sub-agents that operate the CRM safely on the rep's behalf.
- **An autonomous Follow-Up engine.** It ingests inbound emails, builds an **opportunity knowledge graph**, scores deal risk, and drafts the next best action — an email, a note, a task, or a meeting — for the rep to approve.
- **Privacy-first data masking.** Personal data is masked before it ever reaches an external model and unmasked only after the work is complete, so sensitive information never leaves a trusted boundary.

Foundation & architecture at a glance

Interface layer — Twenty CRM (open source) provides the records, UI, and the surfaces where reps give input and approve the agent's proposals. The workflow explicitly pauses for human input and human approval.

Agent layer (Ripple AI) — a planning orchestrator coordinating reader, writer, and Follow-Up sub-agents over a controlled delegation interface, with a shared session and a knowledge graph.

Trust & data layer — PII masking at every model boundary, plus the CRM tools the agents call to read and write records.

WHO IT IS FOR

Ripple is built for **BeamData** and the sales teams it serves. A sales rep's context is scattered across CRM records, emails, notes, tasks and calendars — "data is the new currency, scattered across every tool." That fragmentation has a measurable cost:

75%

of customer information never gets logged into the CRM

\$3–5M

in revenue lost to missing or stale information

121

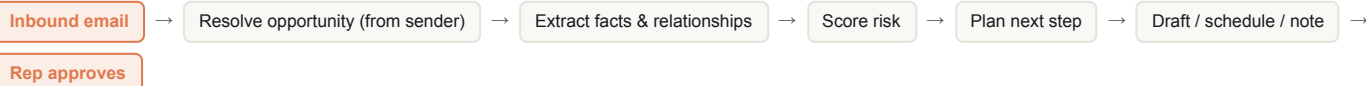
emails reaching a sales rep on an average day

Ripple's job is to capture what would otherwise be lost, keep the deal picture current automatically, and hand the rep a ready-to-approve next step instead of a blank record — without adding manual data entry to their day. The primary user is the **sales rep**; the buyers are **sales leaders** who care about pipeline accuracy, revenue protection, and data privacy.

HOW IT WORKS

Conversational agent (rep-initiated). A user message goes to an **orchestrator** that breaks it into ordered sub-tasks and delegates each to the right specialist: a **reader** (read-only scope) or a **writer** (write scope). The sub-agents call CRM tools — finding records, creating and updating them, and higher-level *composite* actions such as onboarding a client or advancing a deal. Reads return grounded answers; writes are **tiered by risk** — routine changes run directly, while higher-impact actions (such as deletions) pause for **explicit confirmation**.

Follow-Up engine (automatic). An incoming email is first resolved to the right opportunity *from its sender* — the deal is rarely named in the message itself. Key facts and relationships are then extracted, deal risk is scored, and a plan is produced. The pipeline prepares the artifacts (a drafted reply, a calendar availability check, a note or a task) and surfaces a single **pending action** the rep can **Accept, Reject, or Revise**. Nothing is sent or written until the rep accepts.



The opportunity knowledge graph

The knowledge graph is Ripple's core differentiator. It joins **structured CRM records** — owner, contact, company, stage, and deal amount — with **entities surfaced from unstructured text** in notes and emails: champions, blockers, and competing vendors that never had a CRM field of their own. Layered on top is a live **risk score** (for example, "42% chance of loss") with the reasons behind it. The result is a complete, queryable picture of each deal that updates itself as new emails arrive — surfacing, say, that a procurement blocker is pushing back on price while a rival vendor is being preferred, all extracted automatically from the conversation.

Privacy-first masking (the trust zone)

Before any text reaches an external model, Ripple replaces personal data with reversible handles, lets the model reason over the masked version, and restores the real values only at the end — inside a trusted boundary.



Human-in-the-loop where it matters. High-impact and outbound actions never happen on their own — the autonomous Follow-Up engine always proposes and the rep approves, and risky CRM writes pause for confirmation. Routine, low-risk changes run directly to stay out of the rep's way. This graduated control is what makes an autonomous agent safe to put in front of a real sales pipeline.

WHAT AI COMPONENTS WERE USED

- **Multi-agent orchestration on LangGraph.** A planning orchestrator coordinates the reader, writer, and Follow-Up sub-agents through a controlled delegation interface, keeping each agent's scope and permissions separate. The Follow-Up pipeline runs as a directed graph: resolve → extract → load profile → assess risk → plan → run tasks → create pending action.
- **Tool-calling LLM agents.** Large language models (with a configurable backend) select and invoke CRM tools, including composite tools that bundle several CRM operations into one business action. Read and write scopes are enforced per agent.
- **Specialist Follow-Up sub-agents.** A *next-step planner* decides what should happen next, an *email drafter* owns tone and template, and a *risk assessor* reads the deal's facts and returns a calibrated risk score.
- **Knowledge-graph extraction.** LLM-driven extraction of facts, relationships, and provisional ("shadow") entities from unstructured text, with conflict resolution by source priority and entity de-duplication so the same person mentioned two ways is not stored twice.
- **PII detection & masking.** Microsoft Presidio with NER models detects names, emails, and phone numbers and swaps them for reversible handles at every model boundary, then restores them afterwards.
- **Automated evaluation harness.** A scenario-driven, LLM-graded test suite scores the agents on five capabilities — completing the request, using the right tools, handling problems safely, protecting personal data, and staying grounded and accurate — producing a client-facing scorecard.

How Ripple compares

CAPABILITY	RIPPLE	HUBSPOT	MONDAY	MS DYNAMICS 365
AI-assisted sales workflows	✓	✓	✓	✓
Autonomous next-step planning & execution	✓	Limited	✗	Limited
Opportunity knowledge graph	✓	✗	✗	✗
Sensitive-data masking for AI	✓	✗	✗	Limited
Revenue intelligence & risk detection	✓	Limited	✗	✓

WHAT WORKED WELL

- Natural-language read and write against a real CRM, with risk-tiered safeguards and confirmation on high-impact actions.
- Sender-anchored resolution that reliably maps an inbound email to the correct deal even when the deal is never named.
- Building a usable knowledge graph from unstructured text — pulling out champions, blockers, and rivals that had no CRM field.
- PII masking verified end-to-end: no raw names or emails reached the model at the reader boundary in testing.
- The Accept / Reject / Revise loop that keeps the rep firmly in control of every outbound action.
- Cross-turn memory and ambiguity handling ("which Stripe deal did you mean?") performed strongly in evaluation.

WHAT WE WOULD IMPROVE NEXT

- **A researcher agent.** Autonomously enrich deals with external context — company news, key people, funding and market signals — and feed it back into the knowledge graph so the rep walks into every conversation prepared.
- **Higher tool-calling accuracy.** More robust name resolution (including possessives), masking of write-content PII, guaranteed note-to-deal linking, and perseverance through recoverable tool errors instead of giving up early.
- **Scenario coverage to perfection.** Broaden and harden the scenario suite until the agents behave reliably across the full range of real sales situations, using the evaluation scorecard as the gate for what ships.

In short: Ripple already demonstrates the hard parts — autonomous planning, a self-updating knowledge graph, and privacy-safe AI inside a real CRM. The roadmap is about accuracy and reach, not reinvention.

WHAT DID NOT WORK / IS INCOMPLETE

- Tool-calling accuracy gaps: possessive phrasing ("Kevin's") can break name resolution; some newly-written contact details are not yet masked; notes occasionally save without being linked to their deal.
- The orchestrator sometimes gives up after the first tool error instead of attempting a recovery path.
- The risk feature is functional but not yet reflected in the current presentation draft.
- The researcher (external enrichment) agent exists only as a stub.
- Environment and data flakiness — duplicate seed records and backend limits under high concurrency made some evaluation runs harder to interpret.

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